VIGNAN's INSTITUTE OF INFORMATION TECHNOLOGY (AUTONOMOUS)
(Approved by AICTE-New Delhi & Affiliated to JNTU-GV, Vizianagaram)
Beside VSEZ, Duvvada, Vadlapudi Post, Gajuwaka, Visakhapatnam - 530 049.

TITLE:LLM-ASSISTED DISASTER MANAGEMENT CHATBOT USING RETRIEVAL AUGMENTED GENERATION

Abstract:

This project develops an AI-based disaster management chatbot using LLMs and RAG to provide real-time and accurate guidance during disasters.

It enables two-way interaction, supports weather alerts, voice assistance, and multiple languages, and ensures quick, reliable responses.

Introduction:

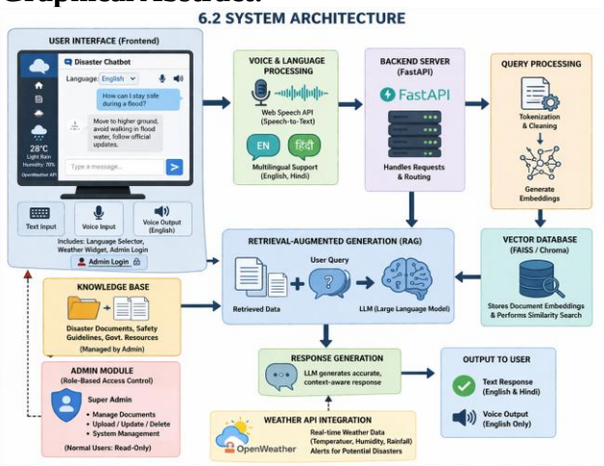
Natural disasters require quick and reliable information, but existing systems provide one-way communication and lack personalization.

This project introduces an AI-based chatbot using LLMs and RAG to deliver real-time, accurate, and user-friendly guidance during emergencies.

Objectives :

- Provide real-time disaster assistance
- Generate accurate responses using RAG
- Support multiple disaster types
- Integrate weather alerts
- Enable voice and multi-language support
- Reduce misinformation using verified data

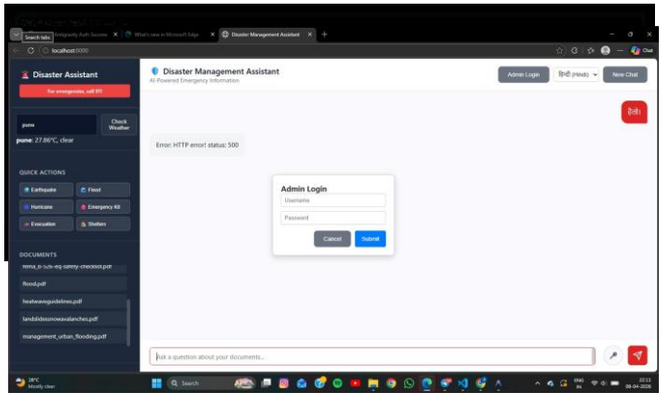
Graphical Abstract:



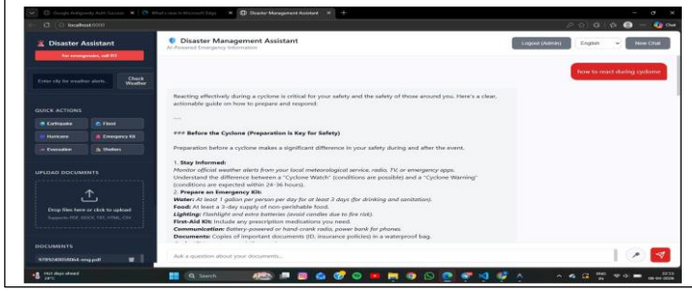
Methodology:

- **Document Processing** – Upload and process disaster-related documents (PDF, DOCX, TXT)
- **Text Chunking & Embedding** – Convert text into vector embeddings
- **Vector Database** – Store embeddings using FAISS/Chroma for fast retrieval
- **Query Processing** – Convert user queries into vectors
- **RAG Framework** – Retrieve relevant information and combine with query
- **Response Generation** – Generate accurate answers using LLM
- **User Interface** – Provide chatbot interface for interaction

Results



Conclusion: This project demonstrates how AI and LLM technologies can transform disaster management systems by providing interactive, real-time, and reliable assistance. The chatbot improves communication during emergencies and ensures users receive accurate safety guidance quickly.

References :. Chen, L., Zhang, Y. (2020), "Chatbot Applications in Emergency Man-agement" , Springer Journal.

Outcomes: